

Symmetry Protection and the Hierarchy Problem: From Lattice Models to Goofy Transformations

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Outline

- 1 Introduction
- 2 Symmetry Protection in the Ising Model
- 3 Mass Corrections in Quantum Field Theory
- 4 Goofy Transformations and the Hierarchy Problem
- 5 Conclusions

The Hierarchy Problem

The Higgs boson is much lighter than the Planck scale.

Quantum corrections to its mass are naively expected to make it comparable to the scale at which new physics appears, unless we perform a **fine-tuning** to cancel these contributions [7].

This is considered one of the main hints of physics beyond the Standard Model.

The question

Can a **symmetry** protect the Higgs mass, the way chiral symmetry protects fermion masses?

Symmetry Protection: the Organizing Principle

't Hooft's naturalness criterion [6]:

A parameter is *naturally small* if setting it to zero **enlarges the symmetry** of the theory.

System	Symmetry	Protected parameter
Ising model	$\mathbb{Z}_2$	odd-spin couplings
Fermion (QFT)	Chiral	fermion mass m_f
Scalar (QFT)	None	—
Higgs (SM)	<i>Goofy?</i>	$m_H^2?$

This thesis: explore symmetry protection in three complementary settings.

The 2D Ising Model with Odd Operators

Extended Hamiltonian:

$$H[\sigma] = - \sum_{d=1}^4 \mathcal{K}_d \sum_{|i-j|=d} \sigma_i \sigma_j - h \sum_i \sigma_i - h_3 \sum_{\langle ijk \rangle} \sigma_i \sigma_j \sigma_k$$

Under $\mathbb{Z}_2$: $\sigma_i \rightarrow -\sigma_i$

- **Even** (symmetry-allowed): $\mathcal{K}_1, \mathcal{K}_2, \mathcal{K}_3, \mathcal{K}_4$
- **Odd** (symmetry-forbidden): h, h_3

Prediction: if $\mathbb{Z}_2$ is exact and $h = h_3 = 0$ microscopically, the RG flow *cannot* generate these operators.

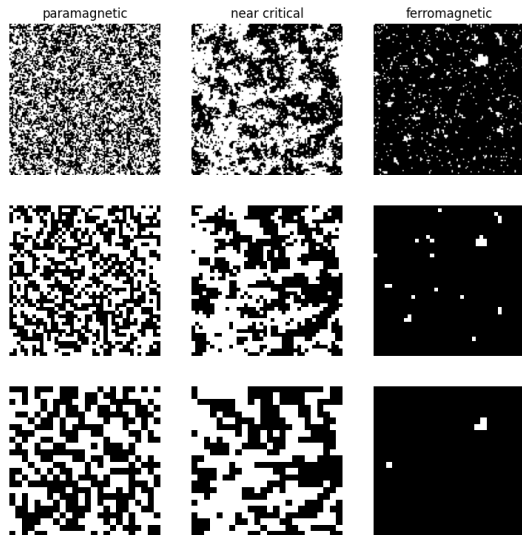
Method: Kadanoff Blocking + Inverse Ising

Pipeline (following Di Carlo [1]):

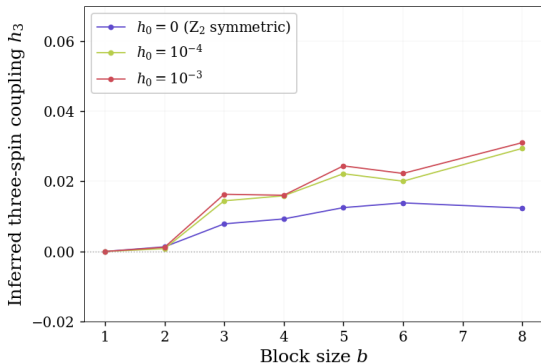
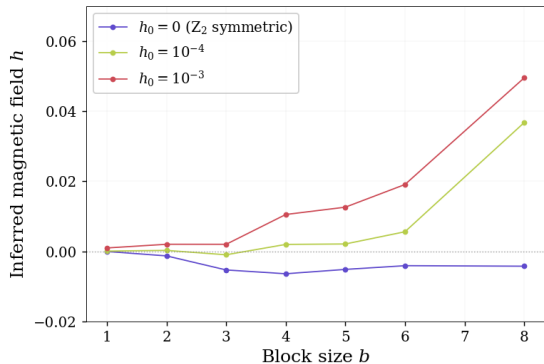
- 1 Monte Carlo sampling ($L=240$, $N_s=300$)
- 2 Kadanoff majority-rule blocking ($b \times b$)
- 3 Pseudo-likelihood inference of $\mathcal{H}^{(b)}$

My contribution:

Extended the inference to include $\mathbb{Z}_2$ -odd operators h and h_3 .

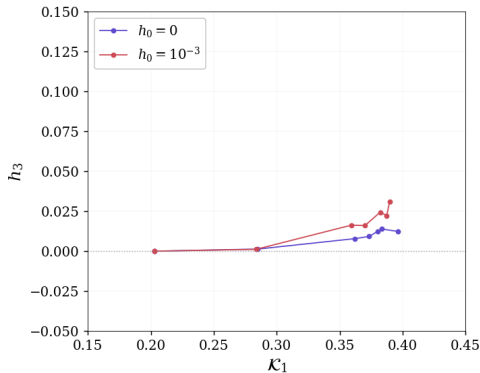
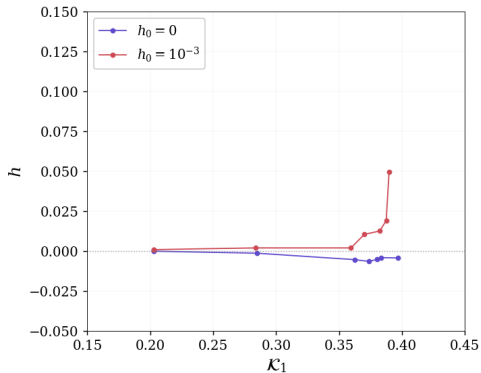


Results: Odd Couplings Under Coarse-Graining



- **Purple** ($h_0=0$): h stays consistent with zero — **protected**.
- **Green / Red** ($h_0 \neq 0$): odd couplings grow — **no protection**.
- h_3 drift in symmetric case: basis-truncation artifact.

RG Flow Trajectories



Symmetric flow: confined near $h, h_3 \approx 0$.

Broken flow: escapes along odd axes.

Ising Model: Takeaway

Key result

An exact symmetry **confines the RG flow** to the symmetric subspace.

Breaking the symmetry, even by a small amount, **opens the entire forbidden sector**.

This is the same mechanism we will now see in quantum field theory.

Fermion Self-Energy: Chiral Protection

Massless Dirac Lagrangian $\rightarrow$ **chiral symmetry**:

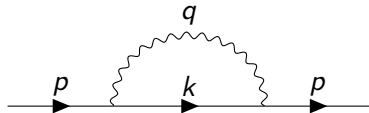
$$\psi \rightarrow e^{i\alpha\gamma_5}\psi$$

The mass term $m\bar{\psi}\psi$ breaks it:

$$m\bar{\psi}e^{2i\alpha\gamma_5}\psi \neq m\bar{\psi}\psi$$

$\Rightarrow$ Setting $m=0$ **enlarges the symmetry**.

Prediction: $\delta m \propto m$.



One-loop fermion self-energy

Fermion Self-Energy: Result

After the full one-loop calculation (dim. reg., on-shell scheme):

$$\delta m = \frac{3g^2 m}{(4\pi)^2} \left[\frac{2}{\varepsilon} + \text{finite} - \ln \frac{m^2}{\mu^2} \right] \propto m$$

In the chiral limit $m \rightarrow 0$: $\delta m \rightarrow 0$. The mass is **protected**.

Scalar Self-Energy: No Protection

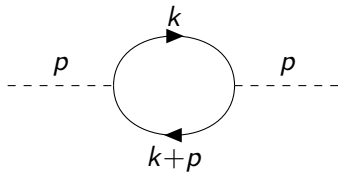
Yukawa theory: $\mathcal{L} \supset -y \phi \bar{\psi} \psi$

The scalar mass operator ϕ^2 is compatible with **all symmetries** of the theory.

$\Rightarrow$ Setting $m_\phi^2=0$ does **not** enlarge the symmetry.

No prediction of protection.

The loop integral will confirm this.



One-loop scalar self-energy
(fermion loop)

Scalar Self-Energy: Result

The scalar mass m_ϕ **never enters the loop**. After integration:

$$\delta m_\phi^2 = \frac{12 y^2 m_\psi^2}{(4\pi)^2} \left[\frac{2}{\varepsilon} + \frac{1}{3} - \gamma_E + \ln(4\pi) - \ln \frac{m_\psi^2}{\mu^2} \right] \propto m_\psi^2$$

- $\delta m_\phi^2 \propto m_\psi^2$, **not** m_ϕ^2 .
- Setting $m_\phi^2 = 0$ does not make the correction vanish.
- The scalar mass is **unprotected**.

The Pattern: Symmetry Protection Across Settings

	Ising	Fermion	Scalar
Symmetry	$\mathbb{Z}_2$	Chiral	None
$m=0$ enlarges sym.?	Yes	Yes	No
$\delta m \propto m$?	N/A	Yes	No ($\propto m_\psi^2$)
Protected?	Yes	Yes	No

The hierarchy problem, restated

The Higgs mass is unprotected because the Standard Model has **no symmetry forbidding** $H^\dagger H$.

Can we *find* such a symmetry?

Origin: the r_0 Symmetry in the 2HDM

Ferreira, Grzadkowski, OGREID & OSLAND (2024) [2]:

In the two-Higgs-doublet model, using bilinear variables:

$$V = M_\mu r^\mu + \Lambda_{\mu\nu} r^\mu r^\nu, \quad r_0 = \frac{1}{2}(\Phi_1^\dagger \Phi_1 + \Phi_2^\dagger \Phi_2)$$

Under $r_0 \rightarrow -r_0$, invariance requires:

$$m_{11}^2 + m_{22}^2 = 0, \quad \lambda_1 = \lambda_2, \quad \lambda_6 = -\lambda_7$$

- These relations are **RG-stable to all orders** in perturbation theory.
- They do not correspond to **any** of the six known 2HDM symmetries.
- The quadratic condition $m_{11}^2 = -m_{22}^2$ is entirely new.

Reinterpretation: $H \rightarrow iH$

Trautner (2025) [3]: the r_0 symmetry is a *field transformation*.

The simplest *goofy* transformation:

$$H \rightarrow iH, \quad H^\dagger \rightarrow iH^\dagger$$

Its defining property — it acts **non-trivially on kinetic terms**:

$$(D_\mu H)^\dagger (D^\mu H) \rightarrow -(D_\mu H)^\dagger (D^\mu H)$$

$$m_H^2 H^\dagger H \rightarrow -m_H^2 H^\dagger H$$

Both built from the same bilinear $\Rightarrow$ both flip sign.

Key consequence

If *goofy* is exact $\Rightarrow m_H^2 = 0$ is enforced, just as chiral symmetry enforces $m_f = 0$.

The Goofy-Symmetric Standard Model

de Boer, Goertz & Incrocci (2025) [4]:

Yukawa sector: $H \rightarrow iH$ introduces a factor of i per Yukawa vertex. Compensated by fermion transformations with $\sqrt{i}$ phases:

$$Q_L \rightarrow \sqrt{i} Q_L, \quad u_R \rightarrow -\sqrt{i} u_R, \quad d_R \rightarrow -\sqrt{i} d_R$$

Kinetic terms: fermion kinetic term also flips: $\bar{\Psi} \not{D} \Psi \rightarrow i \bar{\Psi} \not{D} \Psi$.

Restored by $\partial_\mu \rightarrow -i\partial_\mu$, $A_\mu^a \rightarrow -iA_\mu^a$.

Higgs mass generation:

- Higher-dimensional operators $|H|^{4n}$ (goofy-invariant)
- Extended scalar sector with spontaneous goofy breaking

Mathematical foundation: outer automorphisms constrain β -functions non-perturbatively [5].

Goofy in Context

	Fermion	Higgs (with goofy)
Mass = 0 $\Rightarrow$	enhanced chiral sym.	enhanced goofy sym.
Acts on kinetic term?	trivially	non-trivially
RG-stable?	Yes	Yes (all orders)
Protection mechanism	$\delta m \propto m$	$m_H^2 = 0$ enforced

Goofy is unconventional — it flips kinetic terms and cannot be an exact symmetry of a dynamical theory.

But it **behaves like a symmetry where it matters**: on the RG flow.

Still in its early stages (2024–2026). Many open questions remain.

Conclusions

- ① **Ising model:** extended inverse-Ising inference to $\mathbb{Z}_2$ -odd operators. Showed that exact symmetry confines the RG flow; breaking opens the forbidden sector.
- ② **QFT:** $\delta m_f \propto m_f$ (chiral protection) vs. $\delta m_\phi^2 \propto m_\psi^2$ (no protection). The hierarchy problem can be seen as the **absence of a protecting symmetry** for scalars.
- ③ **Goofy:** a recently proposed transformation that could protect the Higgs mass by acting non-trivially on kinetic terms, with all-order RG stability grounded in outer automorphisms.

Open directions:

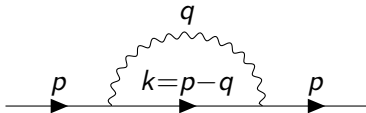
- Lattice realization of a goofy-like symmetry.
- Phenomenological predictions of goofy-symmetric models.

References

- [1] L. Di Carlo, *Am. J. Phys.* **92**, 447 (2024), arXiv:2401.04811.
- [2] P. M. Ferreira et al., *Eur. Phys. J. C* **84**, 234 (2024), arXiv:2306.02410.
- [3] A. Trautner, *Phys. Lett. B* **873**, 140190 (2026), arXiv:2505.00099.
- [4] T. de Boer, F. Goertz, A. Incrocci, arXiv:2507.22111 (2025).
- [5] T. de Boer, A. Trautner, arXiv:2603.12318 (2026).
- [6] N. Craig, *Eur. Phys. J. C* **83**, 825 (2023), arXiv:2205.05708.
- [7] M. E. Peskin, *Nucl. Phys. B* **1018**, 116971 (2025), arXiv:2505.00694.

Backup Slides

Backup: Fermion Self-Energy — Amplitude



Feynman rules: $S_F(k) = \frac{i(\not{k} + m)}{k^2 - m^2}$, $D_{\mu\nu}(q) = \frac{-ig_{\mu\nu}}{q^2}$, vertex: $-ig\gamma^\mu$.

Amputated amplitude:

$$-i\Sigma(p) = -g^2 \int \frac{d^d k}{(2\pi)^d} \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{[k^2 - m^2][(p-k)^2]}$$

Backup: Fermion Self-Energy — Dirac Algebra

d -dimensional identities:

$$\gamma^\mu \gamma^\nu \gamma_\mu = (2 - d) \gamma^\nu, \quad \gamma^\mu \gamma_\mu = d$$

Numerator simplification:

$$\gamma^\mu (\not{k} + m) \gamma_\mu = (2 - d) \not{k} + d m$$

Split into two pieces:

$$-i\Sigma(p) = -g^2 \left[(2 - d) I_k + d m I_1 \right]$$

where $I_k = \int \frac{d^d k}{(2\pi)^d} \frac{\not{k}}{D_1 D_2}$ and $I_1 = \int \frac{d^d k}{(2\pi)^d} \frac{1}{D_1 D_2}$.

The mass renormalization comes **only from the $d m$ piece** $\Rightarrow \delta m \propto m$ is already visible.

Backup: Fermion Self-Energy — Loop Integration

Feynman parametrization:

$$\frac{1}{D_1 D_2} = \int_0^1 dz \frac{1}{[\ell^2 - \Delta]^2}, \quad \ell = k - (1-z)p, \quad \Delta = zm^2 - z(1-z)p^2$$

The vector integral vanishes by symmetric integration:

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{\ell^\mu}{[\ell^2 - \Delta]^2} = 0 \quad \Rightarrow \quad l_k = (1-z)\not{p} \times l_1$$

Standard dim-reg master formula ($d = 4 - \varepsilon$):

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 - \Delta]^2} = \frac{i}{(4\pi)^2} \left[\frac{2}{\varepsilon} - \gamma_E + \ln(4\pi) - \ln \frac{\Delta}{\mu^2} \right]$$

Backup: Fermion Self-Energy — Mass Counterterm

Assembled self-energy:

$$\Sigma(p) = \frac{g^2}{(4\pi)^2} \int_0^1 dz \left[(\varepsilon-2)(1-z)\not{p} + (4-\varepsilon)m \right] \left[\frac{2}{\varepsilon} - \ln \frac{\Delta}{\mu^2} + \dots \right]$$

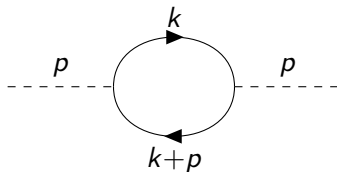
On-shell renormalization: $\delta m = \Sigma(\not{p} = m)$.

Evaluating the z -integral:

$$\delta m = \frac{3g^2 m}{(4\pi)^2} \left[\frac{2}{\varepsilon} + \text{finite} - \ln \frac{m^2}{\mu^2} \right]$$

Every term carries an explicit factor of m : **chiral symmetry in action**.

Backup: Scalar Self-Energy — Amplitude



Yukawa vertex: $-iy$. Closed fermion loop $\Rightarrow$ factor (-1) and trace.

$$i\Sigma_\phi(p^2) = y^2 \int \frac{d^d k}{(2\pi)^d} \frac{\text{Tr}[(\not{k} + m_\psi)(\not{k} + \not{p} + m_\psi)]}{[k^2 - m_\psi^2][(k+p)^2 - m_\psi^2]}$$

Note: m_ϕ does **not** appear anywhere in this expression.

Backup: Scalar Self-Energy — Trace Algebra

Using $\text{Tr}[\gamma^\mu \gamma^\nu] = 4g^{\mu\nu}$, $\text{Tr}[\mathbf{1}] = 4$, odd traces vanish:

$$\text{Tr}[(\not{k} + m_\psi)(\not{k} + \not{p} + m_\psi)] = 4[k \cdot (k+p) + m_\psi^2]$$

Contrast with fermion case:

- Fermion: numerator has Dirac structure $(\not{k}, m)$, mass renormalization comes from the m piece.
- Scalar: trace reduces everything to a **pure scalar function** of k, p, m_ψ .
- No structural reason for m_ϕ to appear $\Rightarrow$ no protection.

Backup: Scalar Self-Energy — Loop Integration

Feynman parametrization and shift $\ell = k + (1-z)p$:

$$\Delta = m_\psi^2 - z(1-z)p^2$$

Numerator $\rightarrow \ell^2 + \Delta = (\ell^2 - \Delta) + 2\Delta$.

Two master integrals:

$$n=1 : \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{\ell^2 - \Delta} = \frac{i \Delta}{(4\pi)^2} \left[\frac{2}{\varepsilon} + 1 - \ln \frac{\Delta}{\mu^2} + \dots \right]$$

$$n=2 : \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 - \Delta]^2} = \frac{i}{(4\pi)^2} \left[\frac{2}{\varepsilon} - \ln \frac{\Delta}{\mu^2} + \dots \right]$$

The $n=1$ integral carries an overall power of Δ — crucial difference.

Backup: Scalar Self-Energy — Mass Counterterm

Assembled self-energy:

$$\Sigma_\phi(p^2) = \frac{4y^2}{(4\pi)^2} \int_0^1 dz \Delta(z, p^2) \left\{ 1 + 3 \left[\frac{2}{\varepsilon} - \ln \frac{\Delta}{\mu^2} + \dots \right] \right\}$$

At $p^2 = 0$: $\Delta \rightarrow m_\psi^2$ for all z . The integral trivializes:

$$\delta m_\phi^2 = \frac{12 y^2 m_\psi^2}{(4\pi)^2} \left[\frac{2}{\varepsilon} + \frac{1}{3} - \gamma_E + \ln(4\pi) - \ln \frac{m_\psi^2}{\mu^2} \right]$$

$\delta m_\phi^2 \propto m_\psi^2$, **not** m_ϕ^2 : the scalar mass is **unprotected**.

Backup: Inverse Ising — Gradients for Odd Operators

Gradient for $\mathcal{K}_d$ (from Di Carlo):

$$\frac{\partial F}{\partial \mathcal{K}_d} = -\frac{2}{N_s L^2} \sum_{t,n} \left(\sum_{|n-m|=d} \sigma_m^{(t)} \right) \frac{\sigma_n^{(t)}}{1 + e^{2\sigma_n^{(t)} H_n^{(t)}}}$$

New contributions (this work):

$$\frac{\partial F}{\partial h} = -\frac{2}{N_s L^2} \sum_{t,n} \frac{\sigma_n^{(t)}}{1 + e^{2\sigma_n^{(t)} H_n^{(t)}}}$$

$$\frac{\partial F}{\partial h_3} = -\frac{2}{N_s L^2} \sum_{t,n} \Theta_n^{(t)} \frac{\sigma_n^{(t)}}{1 + e^{2\sigma_n^{(t)} H_n^{(t)}}}$$

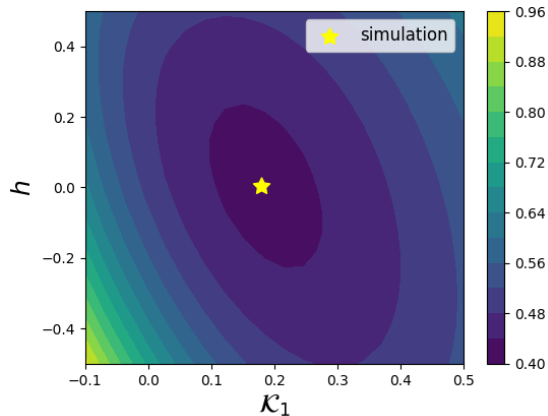
Same structure, different “source fields” ($1, \Theta_n$).

Backup: Inverse Ising — Inference Pipeline

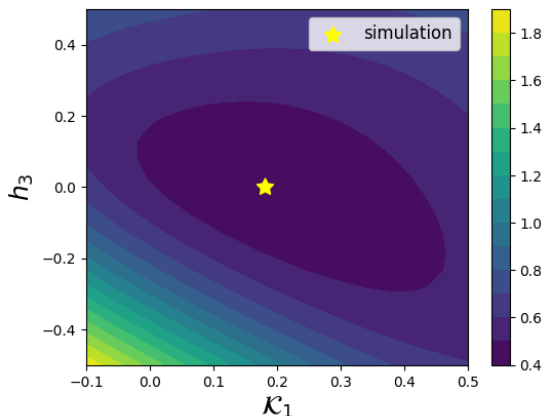
- 1 **Sample:** $N_s = 300$ equilibrium configs on 240×240 lattice (Metropolis, $(\mathcal{K}_1, \mathcal{K}_2) = (0.203, 0.078)$, near-critical)
- 2 **Thermalize:** discard first 100 samples
- 3 **Block:** Kadanoff majority rule for $b \in \{2, 3, 4, 5, 6, 8\}$
- 4 **Infer:** minimize F in the basis $\{\mathcal{K}_1, \dots, \mathcal{K}_4, h, h_3\}$ via gradient descent ($\eta = 0.025$)

Operator basis: truncated to 6 operators. Coarse-graining generates infinitely many — truncation effects discussed next.

Backup: h_3 Drift — Loss Landscape



$(\mathcal{K}_1, h)$: sharp minimum



$(\mathcal{K}_1, h_3)$: elongated contours

The loss surface is **much flatter** in the h_3 direction than in h .
 $\Rightarrow$ Small shifts in h_3 cost very little in the loss.

Backup: h_3 Drift — Why It Doesn't Invalidate Protection

Origin of the drift:

- Blocking generates infinitely many even operators beyond our basis.
- The optimizer absorbs this missing variance into the flattest direction: h_3 .
- This is a **truncation artifact**, not a symmetry violation.

Evidence:

- Symmetric drift: $h_3 \sim 1.4 \times 10^{-2}$ at $b = 6$.
- Broken case: $h_3 \sim 3.1 \times 10^{-2}$ at $b = 8$.
- The **separation between symmetric and broken** trajectories at large b remains clear.

The magnetic field h has a sharp minimum and shows no drift in the symmetric case $\Rightarrow$ the protection of h is unambiguous.

Backup: Goofy — Action on Kinetic and Mass Terms

Under $H \rightarrow iH$, $H^\dagger \rightarrow iH^\dagger$:

$$H^\dagger H \rightarrow (iH^\dagger)(iH) = i^2 H^\dagger H = -H^\dagger H$$

Since $D_\mu H \rightarrow iD_\mu H$:

$$(D_\mu H)^\dagger (D^\mu H) \rightarrow -(D_\mu H)^\dagger (D^\mu H)$$

Both the kinetic and mass terms are built from the same bilinear structure $H^\dagger H$.

$\Rightarrow$ Any transformation that flips one **must** flip the other.

$\Rightarrow$ This is why the *goofy* cannot be an exact symmetry of a dynamical theory (kinetic term $\neq 0$), but can still constrain parameters.

Backup: Goofy — Yukawa Sector Compatibility

Yukawa coupling: $\mathcal{L}_Y \supset -y_d \bar{Q}_L H d_R - y_u \bar{Q}_L \tilde{H} u_R + \text{h.c.}$

$H \rightarrow iH$ introduces a factor of i per Yukawa vertex (two fermion fields).

Compensation: distribute a $\sqrt{i} = e^{i\pi/4}$ per fermion:

$$Q_L \rightarrow \sqrt{i} Q_L, \quad u_R \rightarrow -\sqrt{i} u_R, \quad d_R \rightarrow -\sqrt{i} d_R$$

Each Yukawa term: $(\sqrt{i})^2 \cdot i = i^2 = -1$ per term.

Fermion kinetic term:

$$\bar{\Psi} \not{D} \Psi \rightarrow (\sqrt{i})^2 \bar{\Psi} \not{D} \Psi = i \bar{\Psi} \not{D} \Psi$$

Restored by the spacetime transformation $\partial_\mu \rightarrow -i\partial_\mu$, $A_\mu^a \rightarrow -iA_\mu^a$.

Backup: Goofy — Bilinear Formalism in the 2HDM

Bilinear variables:

$$r_0 = \frac{1}{2}(\Phi_1^\dagger \Phi_1 + \Phi_2^\dagger \Phi_2), \quad \vec{r} = (r_1, r_2, r_3)$$

Potential in Minkowski-like form:

$$V = M_\mu r^\mu + \Lambda_{\mu\nu} r^\mu r^\nu$$

Known symmetries: rotations of $\vec{r}$ (GCP + Higgs-family = 6 classes).

All leave r_0 invariant.

The r_0 symmetry: $r_0 \rightarrow -r_0$.

Invariance $\Rightarrow M_0 = 0, \vec{\Lambda} = 0 \Rightarrow$

$$m_{11}^2 = -m_{22}^2, \quad \lambda_1 = \lambda_2, \quad \lambda_6 = -\lambda_7$$

No doublet transformation produces these relations $\Rightarrow$ genuinely new.

Backup: Simulation Parameters

Parameter	Value
Lattice size	$L = 240$
Boundary conditions	Periodic
Microscopic couplings	$(\mathcal{K}_1, \mathcal{K}_2) = (0.203, 0.078)$ Near-critical regime
Samples per run	$N_s = 300$ (100 discarded for thermalization)
Block sizes	$b \in \{2, 3, 4, 5, 6, 8\}$
Learning rate	$\eta = 0.025$
Operator basis	$\{\mathcal{K}_1, \mathcal{K}_2, \mathcal{K}_3, \mathcal{K}_4, h, h_3\}$
Magnetic field values	$h_0 \in \{0, 10^{-4}, 10^{-3}\}$
Excluded run	$h_0 = 5 \times 10^{-3}$ (convergence failure at large b)